Tadesse K. Bahiru

CURRICULUM VITAE

Ph.D. Student, Department of Computer Science
University of Houston
Houston, TX, USA
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EDUCATION

Ph.D. in Computer Science — Expected Dec 2028
University of Houston, Houston, TX, USA

M.S. in Computer Science — Expected Dec 2026
University of Houston, Houston, TX, USA — Focus: Data Analytics

M.Tech. in Information Technology — May 2018
Parul University, Vadodara, India

B.Sc. in Computer Science — Jul 2015
Wollo University, Dessie, Ethiopia

RESEARCH INTERESTS

- Responsible and accountable AI: fairness, privacy, transparency, and governance.
- Dataset auditing and clinical AI accountability: data quality, bias, and disclosure risk.
- Evaluation of large language models and retrieval-augmented generation systems.

PUBLICATIONS

[1] **T. K. Bahiru** and C. F. Eick, "BiasBridge: Regional Bias Attribution and Source-Conditioned Debiasing for Tabular Data," *2026 IEEE International Conference on Big Data*, Phoenix, AZ, USA, Dec. 14–17, 2026. *Accepted for publication.*

[2] **T. K. Bahiru** and I. A. Kakadiaris, "Codecard: Leveraging LLMs to Evaluate AI Model Code Development with the System Cards Framework," in *Proc. 14th Int. Conf. on Model and Data Engineering (MEDI)*, Cairo, Egypt, 2025, pp. 271–285.

[3] **T. K. Bahiru**, C. Ordonez, and I. A. Kakadiaris, "Data Subcard: Evaluating Privacy, Fairness, Quality, and Protection in Tabular Data, as Part of the System Cards Framework," in *Proc. IEEE Int. Conf. on Data Science and Advanced Analytics (DSAA)*, Birmingham, UK, 2025.

[4] **T. K. Bahiru**, H. Tibebu, and I. A. Kakadiaris, "AI Data Development: A Scorecard for the System Card Framework," in *Proc. Int. Conf. on Information Technology and Artificial Intelligence (ITAI)*, Haryana, India, 2025.

[5] S. B. Haile, T. D. Belay, **T. K. Bahiru**, and T. B. Akele, "Deep Learning-Based Emotion Classification for Amharic Texts," in *Proc. Int. Conf. on Advances of Science and Technology (ICAST)*, 2023, pp. 49–61.

[6] T. B. Akele, **T. K. Bahiru**, T. L. Molla, S. D. Fentie, T. D. Belay, and A. Y. Marye, "Ethiopic Cryptography: An Efficient Hybrid Encryption Algorithm for Reducing Cybersecurity Attack," in *Proc. Int. Conf. on Advances of Science and Technology (ICAST)*, 2023, pp. 269–284.

[7] **T. K. Bahiru**, V. S. Manjula, T. B. Akele, E. A. Tesfaw, and T. D. Belay, "Mining Road Traffic Accident Data for Prediction of Accident Severity," in *Proc. Int. Conf. on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)*, 2023.

[8] **T. K. Bahiru**, D. K. Singh, and E. A. Tessfaw, "Comparative Study on Data Mining Classification Algorithms for Predicting Road Traffic Accident Severity," in *Proc. 2nd Int. Conf. on Inventive Communication and Computational Technologies (ICICCT)*, 2018.

[9] E. A. Tessfaw, B. Ramani, and **T. K. Bahiru**, "Ethiopian Banknote Recognition and Fake Detection Using Support Vector Machine," in *Proc. 2nd Int. Conf. on Inventive Communication and Computational Technologies (ICICCT)*, 2018.

RESEARCH EXPERIENCE

Computational Biomedicine Lab (CBL)

Research Assistant, CRASA Project Jan 2024 – Sep 2025
University of Houston Houston, TX, USA
Community Responsive Algorithms for Social Accountability

- Developed dataset-auditing methods to assess fairness, privacy risk, data quality, and protection in healthcare and other tabular datasets.
- Applied large language models to evaluate AI model code development and documentation through structured assessment frameworks.
- Built reproducible evaluation pipelines, scorecards, and reporting tools to compare datasets and machine learning systems and support AI accountability.

PRESENTATIONS

T. K. Bahiru and I. A. Kakadiaris, "Responsible AI in Healthcare Datasets: Scores for Fairness, Privacy, and Compliance," *Texas Medical Center (TMC) AI Summit*, Houston, TX, USA, February 18, 2025.

TEACHING & MENTORING

Teaching Assistant Sep 2025 – Present
Department of Computer Science, University of Houston Houston, TX, USA

- **Courses supported:** Computer Science and Programming (COSC 1336); Programming and Data Structures (COSC 2436); Data Science II (COSC 4337); Software Design (COSC 6353).

Lecturer Sep 2019 – Dec 2023
Department of Information Technology, Wollo University Dessie, Ethiopia

- Taught Introduction to Artificial Intelligence, Database Systems, and Human-Computer Interaction.
- Designed applied assignments and projects to strengthen students' programming, database, and introductory machine learning skills.
- Supervised senior projects and applied student research in computing and data-oriented topics.

PROFESSIONAL SERVICE

Peer Review

- *IEEE Access* 2025 – Present
- *IEEE International Conference on Biomedical and Health Informatics (BHI)* 2025 – Present

Academic Leadership

Departmental Committee Leadership Dec 2021 – Nov 2023
Wollo University Dessie, Ethiopia

- Led curriculum and departmental coordination to strengthen student preparation for data science and technology careers.
- Organized training sessions and academic initiatives on emerging technologies and professional development.

TECHNICAL SKILLS

Programming	Python, R, SQL, Java, C++
Machine learning	PyTorch, scikit-learn, Hugging Face
LLM / RAG tools	LangChain, ChromaDB, retrieval-augmented generation systems
Data and development	pandas, NumPy, Git/GitHub, Jupyter, Google Colab

AWARDS & HONORS

Graduate Tuition Fellowship (GTF), University of Houston January 2024 - December 2028

Conference Travel Award, Department of Computer Science, University of Houston
IEEE BigData 2026, Phoenix, Arizona.